Ivan Ilin

Machine Learning Researcher | PhD Candidate in Computer Science at KAUST

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Research Profile

Machine learning researcher developing efficient optimization and compression methods for large language models. My work spans pruning, quantization, sparse fine-tuning, pipeline parallelism, distributed optimization, and AI agent collaboration, with an emphasis on methods that connect rigorous analysis to practical open-source systems.

Education

King Abdullah University of Science and Technology (KAUST) — Thuwal, Saudi Arabia
Ph.D. in Computer Science — *Jan 2025–Present*

- Advisor: Peter Richtárik. Advanced to PhD candidacy after successfully defending the proposal *Efficient Optimization and Compression Methods for Large Language Models*, July 2026.

King Abdullah University of Science and Technology (KAUST) — Thuwal, Saudi Arabia
M.S. in Computer Science — *2023–2024*

- Thesis: *Efficient and Fast Pruning of Large Language Models*. Advisor: Peter Richtárik. DOI: 10.25781/KAUST-44481.

Novosibirsk State University — Novosibirsk, Russia
M.Sc. in Automation of Physical and Technical Research — *2021–2023*

GPA: 5.0/5.0.

Novosibirsk State University — Novosibirsk, Russia
B.S. in Automation of Physical and Technical Research — *2017–2021*

GPA: 4.67/5.0.

Research and Professional Experience

Optimization and Machine Learning Laboratory, KAUST — Thuwal, Saudi Arabia
PhD Researcher — *Jan 2025–Present*

- Develop optimization and compression methods for large language models, including block-wise pruning, sparse fine-tuning, and pipeline-parallel training.
- Release reproducible implementations and collaborate on research in LLM quantization and asynchronous distributed optimization.

ExpaSoft — Novosibirsk, Russia
Deep Learning Junior Researcher — *2020–2023*

- Accelerated a Tacotron 2/WaveGlow speech-generation pipeline with structured pruning and distillation; achieved more than 1.5x speedup with improved quality, or 2x speedup with limited quality loss, on an NVIDIA GTX 1050 Ti.

Budker Institute of Nuclear Physics — Novosibirsk, Russia
Undergraduate Research Assistant — *Jan–Aug 2020*

- Built automated tests for Belle II electromagnetic-calorimeter electronics and measured 63% ECL library coverage with gcov.

Grants and Distinctions

KAUST Small Translational Research Grant (sTRG) — $100,000
TRACE Project — *2026–2027*

Transforming Textbooks into Verified, Interactive Courses using Retrieval-Augmented Generation. Primary author of the successful proposal and project team member; lead PI: Peter Richtárik. Award period: July 2026–June 2027.

NeurIPS 2024 Oral: *PV-Tuning: Beyond Straight-Through Estimation for Extreme LLM Compression.*

Teaching Experience

King Abdullah University of Science and Technology — Thuwal, Saudi Arabia
Teaching Assistant, CS 260: Design and Analysis of Algorithms — *Fall 2025; Fall 2026*

Novosibirsk State University — Novosibirsk, Russia
Programming Teacher (Python) — *2022–2023*

Peer-Reviewed Publications

[1] Vladimir Malinovskii, Andrei Panferov, **Ivan Ilin**, Han Guo, Peter Richtárik, and Dan Alistarh. *HIGGS: Pushing the Limits of Large Language Model Quantization via the Linearity Theorem*. NAACL 2025 (Long Papers), pp. 10857–10886. [paper] [doi]

[2] Vladimir Malinovskii, Denis Mazur, **Ivan Ilin**, Denis Kuznedelev, Konstantin Burlachenko, Kai Yi, Dan Alistarh, and Peter Richtárik. *PV-Tuning: Beyond Straight-Through Estimation for Extreme LLM Compression*. NeurIPS 2024 (Oral). [paper] [oral session]

[3] Alexander Tyurin, Marta Pozzi, **Ivan Ilin**, and Peter Richtárik. *Shadowheart SGD: Distributed Asynchronous SGD with Optimal Time Complexity Under Arbitrary Computation and Communication Heterogeneity*. NeurIPS 2024. [paper]

[4] Jihao Xin, **Ivan Ilin**, Shunkang Zhang, Marco Canini, and Peter Richtárik. *Kimad: Adaptive Gradient Compression with Bandwidth Awareness*. 4th Workshop on Distributed Machine Learning (DistributedML 2023), pp. 35–48. [paper]

Preprints and Technical Reports

[1] **Ivan Ilin** and Peter Richtárik. *Local Work, Global Progress: A Convergence Theory of AI Agent Collaboration*. arXiv:2610.28286, 2026. [paper] [code]

[2] **Ivan Ilin** and Peter Richtárik. *QK-Wanda: Coupling Queries and Keys for Unstructured Pruning*. arXiv:2610.01554, 2026. [paper] [code]

[3] **Ivan Ilin**, Philip Zmushko, and Peter Richtárik. *Super-Tuning: From Activation-Aware Pruning to Sparse Fine-Tuning*. arXiv:2607.09287, 2026. [paper] [code]

[4] **Ivan Ilin** and Peter Richtárik. *Demystifying Pipeline Parallelism: First Theory for PipeDream*. arXiv:2606.03498, 2026. [paper] [code]

[5] Abdurakhmon Sadiev, Artavazd Maranjyan, **Ivan Ilin**, and Peter Richtárik. *Ringmaster LMO: Asynchronous Linear Minimization Oracle Momentum Method*. arXiv:2605.18174, 2026. [paper] [code]

[6] **Ivan Ilin** and Peter Richtárik. *Thanos: A Block-wise Pruning Algorithm for Efficient Large Language Model Compression*. arXiv:2504.05346, 2025. [paper] [code]

[7] **Ivan Ilin**. *Hessian of Perplexity for Large Language Models by PyTorch autograd (Open Source)*. arXiv:2504.04520, 2025. [paper]

Selected Research Presentations

PhD Proposal Defense, KAUST — Thuwal, Saudi Arabia
Research proposal defense — *July 2026*
Efficient Optimization and Compression Methods for Large Language Models.

KAUST Computer Science Graduate Seminar — Thuwal, Saudi Arabia
Department-wide research talk — *April 14, 2025*
Efficient Pruning of LLMs. [video]

KAUST Rising Stars in AI Symposium 2025 — Thuwal, Saudi Arabia
Poster presentation — *April 2025*
Thanos: A Block-wise Pruning Algorithm for Efficient Large Language Model Compression.

SANDS Seminar Series, KAUST — Thuwal, Saudi Arabia
Research seminar — *Nov 2024*
Thanos: A Block-wise Pruning Algorithm for Efficient Large Language Model Compression.

Selected Open-Source Software and Outreach

VoiceCut — Open source
Creator and Maintainer — *2025–Present*
Local-first audio and video editor that uses machine learning to identify and remove failed takes, repetitions, false starts, and breaths.

Thanos, SuperTuning, and Randomized PipeDream — Open source
Research Software — *2024–Present*
Reproducible implementations for LLM pruning, sparse fine-tuning, and pipeline-parallel optimization.

Technical Skills

Languages and frameworks: Python, C, C++, PyTorch. **Tools:** Git, Linux/Unix, LaTeX, OpenCV, OpenGL.
Methods: Large language model compression and evaluation; distributed optimization and training.